

ARKit

ARGeoAnchor

Creating a Geo Anchor

convenience init(coordinate: C...

init(name: String, coordinate: ...

convenience init(name: String, ...

Accessing Latitude and Longitude

var coordinate: CLLocationCoo...

Defining Altitude

var altitude: CLLocationDistan...

var altitudeSource: ARGeoAnc...

ARGeoAnchor.AltitudeSource

Multiuser Experiences

ARParticipantAnchor

App Clip Codes

Interacting with App Clip Codes i...

ARAppClipCodeAnchor

Text Annotations

Creating screen annotations for ...

Recognizing and Labeling Arbitr...

Common Types

ARAnchor

ARAnchorCopying

Filter

/

ARKit / ARGeoAnchor

Class

ARGeoAnchor

An anchor that identifies a geographic location using latitude, longitude, and altitude data.

iOS 14.0+ | iPadOS 14.0+

class ARGeoAnchor

Overview

A *geographic anchor* (also known as *location anchor*) identifies a specific area in the world that the app can refer to in an AR experience. As a user moves around the scene, the session updates a location anchor's [transform](#) based on the anchor's [coordinate](#) and the device's compass heading.

ARKit aligns location anchors to an East-North-Up orientation, with its x- and z-axes matching the longitude and latitude directions. For more information, see [ARConfiguration.WorldAlignment.gravityAndHeading](#).

ARKit sets the anchor's vertical position to the altitude you pass in to [initWithCoordinate:altitude:](#). If you initialize a location anchor using [initWithCoordinate:](#) instead, ARKit sets the anchor's altitude to ground level.

Important

Location anchors are available only in geotracking sessions, and geotracking is available in specific areas; for more information, see [ARGeoTrackingConfiguration](#).

Communicate Data Usage

Location anchors consume data from Apple Maps called *localization imagery* (for more information, see [ARGeoTrackingConfiguration](#)). As the user moves, the framework downloads localization imagery to refine the user's precise geographic position. The amount of data the session requires depends on the user's movement and distance they travel. To make the user aware of potential fees, you can notify the user of their data usage.

Manage Location Anchor Availability

When an app creates a location anchor, it's invisible to the user until the framework *populates* the anchor in the scene. When an anchor populates successfully, the session passes the anchor into the delegate's [session\(\\_:didAdd:\)](#) callback.

If ARKit fails to populate a location anchor, the session calls [session\(\\_:didRemove:\)](#) to notify your delegate. A location anchor may fail to populate when:

- The network is unavailable. If you create a location anchor without providing an altitude, ARKit defaults the altitude to ground level, and may query the server to check the topography at the anchor's geographic coordinate. If the network is unavailable, ask the user to restore a connection by disabling Airplane Mode, enabling WiFi, or moving to a location that provides service. If the network is available but slow, an altitude query response may be delayed. Consider pausing a navigation or presenting visual feedback for the anchor's tentative placement, such as by displaying a status indicator.
- The location anchor is too far from the user. If users can create location anchors in your app, let them know to position the coordinates of each anchor within 0.05 degrees (~5 kilometers) of themselves and their device.
- The server prevents the location anchor's position, such as in a large body of water.

Topics

Creating a Geo Anchor

convenience [init\(coordinate: CLLocationCoordinate2D, altitude: CLLocationDistance?\)](#)

Initializes a location anchor with the given coordinate and altitude.

[init\(name: String, coordinate: CLLocationCoordinate2D, altitude: CLLocationDistance\)](#)

Initializes a named location anchor with the given coordinates and altitude.

convenience [init\(name: String, coordinate: CLLocationCoordinate2D, altitude: CLLocationDistance?\)](#)

Initializes a named location anchor with the given coordinates and altitude.

Accessing Latitude and Longitude

var [coordinate: CLLocationCoordinate2D](#)

The latitude and longitude of the anchor's geographic location.

Defining Altitude

var [altitude: CLLocationDistance?](#)

Vertical distance, in meters, between this anchor and sea level.

var [altitudeSource: ARGeoAnchor.AltitudeSource](#)

A record of the source from which an altitude came.

enum [AltitudeSource](#)

Options for setting a location anchor's altitude.

Relationships

Inherits From

[ARAnchor](#)

Conforms To

[ARAnchorCopying](#)  
[ARTrackable](#)  
[CVarArg](#)  
[CustomDebugStringConvertible](#)  
[CustomStringConvertible](#)  
[Equatable](#)  
[Hashable](#)  
[NSCoding](#)  
[NSCopying](#)  
[NSObjectProtocol](#)  
[NSSecureCoding](#)  
[Sendable](#)  
[SendableMetatype](#)

See Also

Geotracking

[Tracking geographic locations in AR](#)

Track specific geographic areas of interest and render them in an AR experience.

Platforms

iOS

iPadOS

macOS

tvOS

visionOS

watchOS

Tools

Swift

SwiftUI

Swift Playground

TestFlight

Xcode

Xcode Cloud

SF Symbols

Topics & Technologies

Accessibility

Accessories

App Extension

App Store

Audio & Video

Augmented Reality

Design

Distribution

Education

Fonts

Games

Health & Fitness

In-App Purchase

Localization

Maps & Location

Machine Learning & AI

Open Source

Security

Safari & Web

Resources

Documentation

Tutorials

Downloads

Forums

Videos

Support

Support Articles

Contact Us

Bug Reporting

System Status

Account

Apple Developer

App Store Connect

Certificates, IDs, & Profiles

Feedback Assistant

Programs

Apple Developer Program

Apple Developer Enterprise Program

App Store Small Business Program

MFI Program

News Partner Program

Video Partner Program

Security Bounty Program

Security Research Device Program

Events

Meet with Apple

Apple Developer Centers

App Store Awards

Apple Design Awards

Apple Developer Academies

WWDC